

6CS012 - Artificial Intelligence and Machine Learning. Module Handbook and Syllabus.

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1 Basic Course Information.

- **Module Leader:** Siman Giri {siman.giri@herald}.
 - **Faculty Members:**
 - Sunita Parajuli - Associate Module Leader.
 - Durga Pokhrel.
 - Shiv Kumar Yadav - In-charge for Course Logistic.
 - Anish Khatiwada.
 - Bijay Ghimire.
 - Suren Tamang - Teaching Assistant
 - **Lectures:** Follow the Schedule.
 - **Office Hours:** On Demand and By Appointment.
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 - Course Logistics and Learning Outcomes.
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2 Course Logistics.

2.1 Course Overview:

This is the second course in **Artificial Intelligence (AI) and Machine Learning (ML)**, building on foundational concepts to explore advanced **deep learning methodologies** and their **theoretical underpinnings**. The course focuses on applications in **computer vision and natural language processing** while reinforcing essential mathematical foundations, particularly in **linear algebra and matrix operations**, which are critical for understanding modern ML architectures.

The course covers key deep learning techniques, including:

- **Convolutional Neural Networks (CNNs)** for feature extraction and image processing.
- **Transfer Learning** for leveraging pre-trained models in new learning tasks.
- **Text Encoding Methods** to represent and process textual data.
- **Recurrent Neural Networks (RNNs)** and their variants for sequential and temporal data modeling.
- **Autoencoders** for dimensionality reduction and representation learning.
- **Transformers and Attention Mechanisms**, foundational to state-of-the-art natural language processing (NLP) and vision models.

A central objective of the course is to develop a **critical understanding of model design, evaluation, and interpretability**. Students will examine the impact of **data preprocessing, feature selection, and hyperparameter tuning** on model performance. Furthermore, the course emphasizes the importance of **reproducibility, scalability, and ethical considerations** in deploying AI solutions.

By the end of the course, students will have gained the necessary skills to analyze **complex unstructured datasets**, **design and implement deep learning models**, and **critically evaluate their effectiveness** in real-world applications.

2.2 Pre-requisites:

Must have obtained passing grades on following Modules:

- Introduction to Python Programming - Level - 4.
- Computational Mathematics - Level - 4.
- 5CS037 - Concepts and Technologies of AI - Level - 5.
- 6CS005 - High Performance Computing - Level - 6.

2.3 Recommended Reading:

There are no required textbooks for this course. Lecture slides, notes, and worksheets will be sufficient for covering the course material. If additional reading materials or research papers are needed, they will be provided as necessary. However, the following resources are recommended to further support your learning:

- **Ian Goodfellow, Yoshua Bengio, and Aaron Courville, Deep Learning, MIT Press, 2016.**
- **Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.**
- **François Chollet, Deep Learning with Python, Manning Publications, 2021 (2nd Edition).**
- **Hands-On Machine Learning With Scikit-Learn, Keras, and Tensorflow, O'Reilly, Aurélien Géron, Latest edition 2019.**

2.4 Learning Outcomes:

Upon successful completion of this course, students will be able to:

1. Understand Key Concepts in Deep Learning:
 - Demonstrate a solid understanding of the fundamental issues and challenges in machine learning, including data quality, model selection, overfitting, and model complexity.
 - Analyze the strengths and weaknesses of various deep learning approaches.
2. Develop Theoretical and Mathematical Foundations:
 - Appreciate the underlying mathematical relationships within and across deep learning models, particularly in the context of supervised and unsupervised learning paradigms.
 - Apply concepts from linear algebra, matrix operations, and optimization to understand and improve deep learning architectures.
3. Apply Deep Learning Techniques to Real-World Problems:
 - Identify and select appropriate deep learning models, such as CNNs, RNNs, Autoencoders, and Transformers, based on the problem domain (e.g., computer vision, natural language processing).
 - Implement state-of-the-art deep learning algorithms using modern frameworks and evaluate their performance on real-world datasets.
4. Develop Practical Skills for Model Design and Evaluation:
 - Design and optimize deep learning models, considering factors such as data preprocessing, feature selection, hyperparameter tuning, and interpretability.
 - Assess and compare model performance using appropriate evaluation metrics.

3 Final Grading:

The requirements for this course include active participation in class discussions, completion of assigned lab exercises, a mid-term quiz, and the final course project. The grading breakdown is as follows:

- **20% Mid - term Quiz:** A quiz in MCQ format, Date and Time to be announced.
- **10% Weekly Worksheets, Problem Sets and Question Answer.**
- **60% Final Course Project:** The project will be divided into two components as mentioned below:
 1. **Project - I - Computer Vision Task.**
 2. **Project - II - Natural Language Processing Task.**
- **5% Report Writing:** Quality of Report presented.
- **5% Participation:** Active participation will be assessed through in-class quizzes and involvement in class discussions. Please note that no make-up quizzes will be provided for missed classes.

This grading structure is designed to encourage consistent engagement with the material and foster both individual and collaborative learning throughout the course.

Caution: These plans are tentative and subject to change. Final confirmation will be provided upon approval from the university.

4 Getting Help:

The preferred method of communication for course-related inquiries is via email, ensuring a timely response. For urgent matters or recommendations, students are encouraged to meet with the teaching staff or the designated teaching assistant (TA) during their office hours.

4.1 About Office Hours:

The teaching staff holds weekly office hours to assist students with course-related matters. Students can obtain details of office hours time by contacting their **designated instructor**, the **TA**, the **Student Support Desk (SSD)**, or the **Personal Academic Tutor (PAT) office**. Meetings outside of scheduled office hours are strictly by prior **appointment only**.

4.2 Designated Teaching Assistant (TA):

A designated teaching assistant (TA) has been assigned to this module as the primary point of contact for course-related inquiries. However, students should note that TAs have additional responsibilities and may only be available during office hours, via email, or by appointment.

Teaching Assistant for this Module - Suren Tamang.

5 Academic Integrity Policies:

The Academic Integrity Policy of this course must be strictly adhered to when completing assignments and participating in discussions.

Please - Read this carefully.

5.1 Collaboration among Students:

Collaboration among students is intended to **facilitate deeper learning and comprehension, rather than to circumvent the learning process**. Engaging in group discussions and collaborative study of course materials is strongly encouraged. Students may seek clarification and conceptual guidance from their peers to enhance their understanding of the subject matter required for completing assignments. However, such collaboration must support independent learning rather than substitute for individual effort. **The direct reproduction of any material from other students or external sources is strictly prohibited**. All submitted work must be the sole effort of each student.

All of the following activities will be considered cheating:

1. **Sharing or Copying code, files or answers:** whether through direct copying, retyping or using online sources e.g. Stack-overflow Git-hub without proper attribution.
2. **Submitting work that is not original:** including using external code or code generated by LLMs, with intention of passing off such work as the student's own.
3. **Copying answers** to quizzes, assignments, or projects from another individual or from any published or unpublished written or electronic sources.
4. **Collaborating with others** on individual assignments, quizzes or project without explicit permission from the instructor.

5.2 Duty to Protect One's Work:

Students are responsible for safeguarding their work against unauthorized access, copying, or misuse by others. If a student's work is copied by another student, both parties will be held accountable for violating course policies. This applies regardless of whether the original author explicitly permitted the copying or failed to take adequate precautions to prevent it. If identical or highly similar work is submitted by multiple students, all involved will face academic penalties.

To uphold academic integrity and protect future students, solutions to assignments must not be shared publicly, either during the course or after its completion.

5.3 Duty to Give Viva:

It is **student's responsibility to attend viva** examinations at the scheduled time determined by the instructor. Failure to comply with this requirement may result in academic penalties.

5.4 Plagiarism and AI - Generated Content:

Plagiarism, including the use of AI-generated content without proper attribution, constitutes a violation of academic integrity. Students must ensure that all work submitted for evaluation is their own and properly cites any external sources, including AI-generated material, used in their assignments.

Using AI tools to generate content without disclosing their use, or presenting AI-generated work as one's own, is considered plagiarism. Any attempt to submit plagiarized work, whether through manual copying or the use of AI tools, will be subject to academic penalties. It is essential that students recognize the importance of maintaining transparency regarding the sources and tools utilized in their work to uphold academic standards.

5.5 Other General Policy:

Students are expected to **attend all lectures, tutorials, and workshops** regularly. **Active participation** in class discussions is encouraged, and students must ensure the timely submission of all coursework as assigned by the instructor. While late submissions may be accepted under certain circumstances, they will incur penalties as per the course policy.

In addition to the aforementioned policies, students are expected to adhere to all rules and regulations established by the College . **Failure to comply with these policies may result in severe academic penalties, including potential exclusion from the module.**

6 Approximate Schedule

This is a breakdown of the topics we will cover every week and are tentative and may vary in actual.

Week - 1 - Review of Matrices and Derivatives for Deep Learning:

- **Lecture:** Briefly review the concepts of vectors, matrices, and derivatives, and extend these ideas to discuss gradients. Focus on how gradients are used to find the maxima or minima of functions. Additionally, introduce matrix decomposition with practical applications, including Principal Component Analysis (PCA).
- **Tutorial:** Hands-on exercises focused on applying gradients to find maxima and minima, as well as implementing PCA for dimensionality reduction.
- **Workshop:** Implement PCA for image compression, showcasing its application in data reduction.

By the end of this week, students must complete and submit:

Worksheet - I on PCA for Image Compression.

Week - 2 - Image Classification with Linear Classifiers:

- **Lecture:** Introduce the fundamental learning paradigms, including supervised and unsupervised learning, and provide an overview of common learning frameworks. Emphasize the principles of linear classification, with a particular focus on logistic regression as a foundational method for image classification. Situate logistic regression within the Empirical Risk Minimization (ERM) framework, highlighting its theoretical underpinnings and practical applications. Additionally, introduce gradient descent as a fundamental optimization technique, explaining its role in minimizing loss functions and improving model performance.
- **Tutorial:** Examine key challenges in image classification, such as variations in lighting, scale, and orientation, as well as the high-dimensional nature of image data. Students will develop a logistic regression-based classifier, progressing through the formulation of the hypothesis function, cost function, and gradient derivation, followed by parameter updates using optimization techniques. A pen-and-paper exercise will reinforce the theoretical foundations, guiding students through the process to facilitate coding implementation in the workshop. Finally, students will critically examine the limitations of logistic regression in handling complex, non-linear decision boundaries, setting the stage for exploring more advanced models like neural networks.
- **Workshop:** Implement Softmax Regression for multi-class classification on the MNIST dataset, training the model using cross-entropy loss and gradient descent. They will analyze the model's performance and identify its limitations, particularly its struggle with complex, non-linear patterns. This will lead to a discussion on the need for neural networks, demonstrating how Softmax Regression's inability to capture hierarchical features highlights the necessity of deeper architectures for improved classification.

By the end of this week, students must complete and submit:

Worksheet -II on MNIST Image Classification with Logistic Regression.

Week - 3 - Getting Started with Neural Networks: From Neurons to Single Layer and Multilayer Perceptrons:

- **Lecture:** The introduction to neural networks begins with an exploration of their historical evolution, tracing their roots back to early models of artificial neurons and their inspiration from biological systems. This section will provide a foundational understanding of how neurons function within a neural network and how this concept directly connects to logistic regression. We will examine the perceptron learning theory, which serves as a fundamental building block for neural network development, and explore the mechanisms by which a single-layer network is trained to classify linearly separable data. Furthermore, we will delve into feedforward networks, discussing their architecture and the flow of information through multiple layers. We will also critically assess the limitations of feedforward networks, particularly in their inability to model complex, non-linear decision boundaries, thereby motivating the need for more advanced neural network architectures, such as multilayer perceptrons, that can handle such complexity.
- **Tutorial:** Students will engage in hands-on exercises focused on perceptron learning and feedforward networks. The session will cover the fundamentals of perceptron training, including weight updates, activation functions, and error correction. Additionally, students will explore feedforward networks, understanding how multi-layer structures process information and their limitations in handling complex data patterns.
- **Workshop:** Students will implement a feedforward network, applying theoretical concepts to build a functional model for classification tasks. The session will critically examine the network's limitations in modeling nonlinear decision boundaries, highlighting the need for multilayer networks to address complex classification challenges.

By the end of this week student must complete and submit:

Worksheet - III - Code the very basic Perceptron and realize the issues with Linearly separable and Non linearly separable cases.

Week - 4 - Training and Optimization techniques for Multilayer Neural Network:

- **Lecture:** The lecture introduces the foundational concepts of multilayer perceptrons (MLPs), including their structure, training mechanisms, and key components such as backpropagation. It also covers popular gradient descent variants—Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent — emphasizing their advantages, such as faster convergence and noise handling, and their practical applications in deep learning. Additionally, the role of loss functions in training is discussed, with particular focus on Cross-Entropy Loss, a critical function for classification tasks. The relationship between loss functions, optimization methods, and the Empirical Risk Minimization (ERM) framework is explored to provide a thorough understanding of MLP training.
- **Tutorial:** In the tutorial, students engage in pen-and-paper exercises designed to deepen their understanding of backpropagation. They will work through examples of the weight update process in multilayer networks, calculating gradients for different layers and understanding how errors propagate backward through the network. This hands-on approach ensures a solid grasp of the mathematical principles underlying neural network training.

- **Workshop:** In the workshop, students apply their theoretical knowledge by implementing a multi-layer neural network using the Keras framework. They will become familiar with Keras' high-level API for building and training deep learning models, including model compilation, loss functions, and optimization algorithms. Students will train their networks on a practical dataset, gaining experience with the training process and exploring the effects of different optimization techniques on model performance.

By the end of this week student must complete and submit:

Worksheet - IV - Multilayer Perceptron for Classification of Devnagari Handwritten Digits.

Week - 5 - Introduction to Convolutional Neural Networks (CNNs) for Image Classification:

- **Lecture:** The lecture provides a comprehensive overview of Convolutional Neural Networks (CNNs) and their application in image classification. Key topics include the structure of CNNs, which consists of convolutional layers, pooling layers, and fully connected layers. The lecture will explore how CNNs automatically learn hierarchical patterns in images, enabling them to perform effectively in tasks such as object detection and recognition. Additionally, the concepts of feature maps, filters, and the role of stride and padding in convolutional operations will be discussed, highlighting their importance in controlling the output size and capturing spatial relationships within images. This foundational knowledge equips students with a solid understanding of CNNs' capabilities and applications in deep learning.
- **Tutorial:** In the tutorial, students will engage in hands-on exercises to understand the architecture and functioning of Convolutional Neural Networks (CNNs). They will explore the role of convolutional layers in detecting image patterns, pooling layers in reducing spatial dimensions, and fully connected layers in classification. Through a combination of pen-and-paper exercises and simple code implementations, students will visualize intermediate outputs and manually compute convolution operations, enhancing their understanding of how CNNs extract hierarchical features for image classification and recognition tasks.
- **Workshop:** In the workshop, students will implement a Convolutional Neural Network for image classification using Keras. They will apply their theoretical knowledge to build a CNN model from scratch, covering the entire pipeline from preprocessing image data to model training and evaluation. Students will use Keras' high-level API to define the CNN architecture, compile the model, and train it on a benchmark dataset.

By the end of this week student must complete and submit:

Worksheet - V - Image Classification with CNN.

Week - 6 - Enhancing Convolutional Neural Networks: Techniques and Optimization Strategies

- **Lecture:** The lecture will delve into essential practical techniques for optimizing Convolutional Neural Networks (CNNs) in real-world applications. Topics covered will include data augmentation, which enhances model robustness by artificially increasing the diversity of the training dataset; dropout, a regularization method to prevent overfitting by randomly deactivating units during training; and batch normalization, a technique used to normalize the input of each layer to improve convergence and model stability. The lecture will also introduce transfer learning, allowing students to fine-tune pre-trained models on new datasets, leveraging prior knowledge for faster and more effective learning.
- **Tutorial:** In the tutorial, students will investigate regularization techniques, including dropout and batch normalization, and assess their impact on CNN performance. Through hands-on exercises, they will implement these methods to enhance generalization, mitigate overfitting, and improve model robustness. The tutorial will guide students in evaluating the effects of these techniques on training dynamics, validation accuracy, and convergence rates, thereby deepening their understanding of their practical application in deep learning models.
- **Workshop:** In the workshop, students will implement a CNN using a pre-trained network and apply transfer learning for a specific image classification task. They will fine-tune the model by modifying layers, optimizing hyperparameters, and incorporating regularization techniques to improve accuracy and efficiency. This session will provide hands-on experience with advanced CNN architectures, preparing students to develop high-performance models for real-world applications.

By the end of this week student must complete and submit:

Worksheet - VI - Implementing and Fine-Tuning CNNs with Regularization Techniques.

Release of Assignment - I - Image Classification.

Week - 7 - Autoencoders: An Encoder-Decoder Architecture for Learning Latent Features:

- **Lecture:** The lecture provides an in-depth introduction to autoencoders, focusing on their role in unsupervised learning for tasks such as dimensionality reduction and noise removal. It covers the architecture of autoencoders, detailing the encoder and decoder networks, and how they learn compact, efficient latent representations of data. This lecture also explores the concept of training unsupervised deep learning models, with a focus on autoencoders, which are trained without labeled data. Students will gain an understanding of how unsupervised learning is achieved through the minimization of reconstruction error, allowing the model to learn meaningful features and compress data into a lower-dimensional latent space.
- **Tutorial:** In this tutorial, students will train an autoencoder on an image dataset, focusing on how the model learns to compress image data into a lower-dimensional latent space, preserving critical features for reconstruction. The tutorial will emphasize how the autoencoder's encoder-decoder architecture enables the reduction of image data into compact representations while minimizing

information loss. Students will gain hands-on experience in using these compressed latent features for tasks such as image compression and noise removal, understanding how autoencoders can effectively enhance image quality and reduce dimensionality for more efficient storage and processing.

- **Workshop:** This workshop will serve as a revision session for all material covered up until this point in the course. It will also function as a feedback session for Assignment 1 (Image Classification), providing students with the opportunity to review their progress, seek clarification, and receive guidance on key aspects such as data preparation, model implementation, and evaluation. Students will have the chance to address any challenges they may be facing with their assignments and confirm the correctness of their approaches before proceeding further.

No Worksheet to Submit this week.

Week - 8 - Text Preprocessing and Representation for NLP: From Tokenizations to Word Embeddings

- **Lecture:** This session covers essential text preprocessing techniques for natural language processing (NLP), including tokenization, stemming, lemmatization, and text normalization. The lecture will explore text representation methods like TF-IDF and word embeddings (e.g., Word2Vec), focusing on their role in converting raw text into meaningful numerical features for machine learning. Additionally, the concept of word embeddings will be examined, highlighting their ability to capture semantic relationships, such as word analogies (e.g., “king - man + woman = queen”), and their importance in preparing text for NLP tasks.
- **Tutorial:** In this session, students will implement text preprocessing techniques using Python libraries like NLTK. They will focus on text cleaning, normalization, tokenization, stemming, and lemmatization. The tutorial will also include a hands-on demonstration of Word2Vec for generating word embeddings and exploring semantic relationships. Practical exercises will reinforce the lecture’s theoretical concepts and provide experience in transforming text data for NLP tasks.
- **Workshop:** In this workshop, students will apply the text preprocessing techniques learned in the lecture and tutorial to build a Naive Bayes classifier for text classification. They will experiment with various text embeddings, including TF-IDF and Word2Vec, and assess their impact on model performance. This hands-on experience will deepen students’ understanding of machine learning model implementation and the comparative effectiveness of different text representation methods in NLP tasks.

By the end of this week student will complete and submit:

Worksheet - VII - Text Classification with Naive Bayes Classifier.

Release of Assignment - II - Question and Answer.

Week - 9 - Introduction to RNNs for Sequence-to-Sequence Text Modeling and Understanding:

- **Lecture:** This session offers a comprehensive introduction to Recurrent Neural Networks (RNNs), emphasizing their application in sequence modeling and text comprehension tasks. The lecture covers the fundamental architecture of RNNs, highlighting their ability to process sequential data by maintaining hidden states across time steps. It also addresses key challenges such as vanishing gradients, which hinder the network's ability to capture long-term dependencies in sequences. To overcome these limitations, the session introduces Long Short-Term Memory (LSTM) networks, a powerful variant of RNNs designed to better capture long-range dependencies and improve performance in sequence-based tasks, including natural language processing. The session will emphasize the importance of RNNs and LSTMs in tasks that require understanding context and relationships across sequences, such as sentiment analysis and language modeling.
- **Tutorial:** In this tutorial, students will learn how to structure a Recurrent Neural Network (RNN) for a language translation task using a sequence-to-sequence model. The tutorial will guide students through designing the encoder-decoder architecture, where the encoder processes the input sequence and the decoder generates the translated output. Students will also explore the training and evaluation process, focusing on data preprocessing, model optimization, and performance metrics. Through hands-on exercises, students will gain a practical understanding of machine translation models and their real-world applications.
- **Workshop:** In this workshop, students will implement a language translation task using Long Short-Term Memory (LSTM) networks. They will build an LSTM-based sequence-to-sequence model, applying it to translate text from one language to another. The workshop will also focus on understanding the limitations of traditional RNNs, such as vanishing gradients and difficulty in capturing long-term dependencies, and how LSTMs overcome these challenges through their gating mechanisms. Students will gain practical experience in training LSTM models for language translation and analyzing their performance. By the end of the workshop, students will have a comprehensive understanding of how to build, evaluate, and improve language translation models using LSTM networks, as well as how to address common challenges faced in machine translation tasks.

By the end of this week student will complete and submit:

Worksheet - VIII - Machine Translation with RNN and LSTM.

Release of Assignment - III - Text Classification.

Week - 10 - Understanding Transformer Architecture and Attention Mechanisms in NLP:

- **Lecture:** This week introduces the Transformer architecture, the foundation for modern NLP models such as BERT and GPT. The lecture covers key concepts, including the structure of Transformers, self-attention and multi-head attention mechanisms, and positional encoding, which enable the model to focus on different parts of the input sequence and capture complex dependencies. The session also includes a comparative analysis of RNN-based models and Transformers, emphasizing the advantages of Transformers for sequence-based NLP tasks. By understanding these mechanisms, students will gain insight into how Transformers significantly improve performance in handling long-range dependencies and contextual information in text.
- **Tutorial:** In this tutorial, students will implement a Transformer model for text classification tasks, gaining hands-on experience in constructing and training the model. They will explore how the self-attention and multi-head attention mechanisms enable the model to focus on important parts of the input sequence, thereby enhancing its ability to process long-range dependencies in text. Through training and evaluation, students will learn how to fine-tune the model's hyperparameters, optimize the attention mechanism, and assess the performance of Transformer-based models for NLP tasks. This exercise will deepen their understanding of the Transformer architecture's advantages over traditional models and provide practical experience in optimizing attention mechanisms for improved performance.
- **Workshop:** In this workshop, students will review key concepts from the course, with a focus on neural networks, deep learning, and NLP. Feedback on assignments 2 and 3 will be provided, along with guidance for the upcoming exam. The session aims to reinforce understanding, clarify doubts, and ensure readiness for the final assessment.

No Worksheet to Submit this week.

Week - 11 - Final Exam:

About Exam:

A 90 mins. Multiple Choice Quiz on Canvas covering the material from Week 1 to Week 9.

Week - 12 - Final Viva:

About Viva:

A viva on your final portfolio which is the combination of all three assignments.

————— Good Luck —————

————— The - End —————